```
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_digits
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import silhouette_score
from sklearn.metrics import adjusted_rand_score
```

```
digits = load_digits()

X = digits.data
y = digits.target

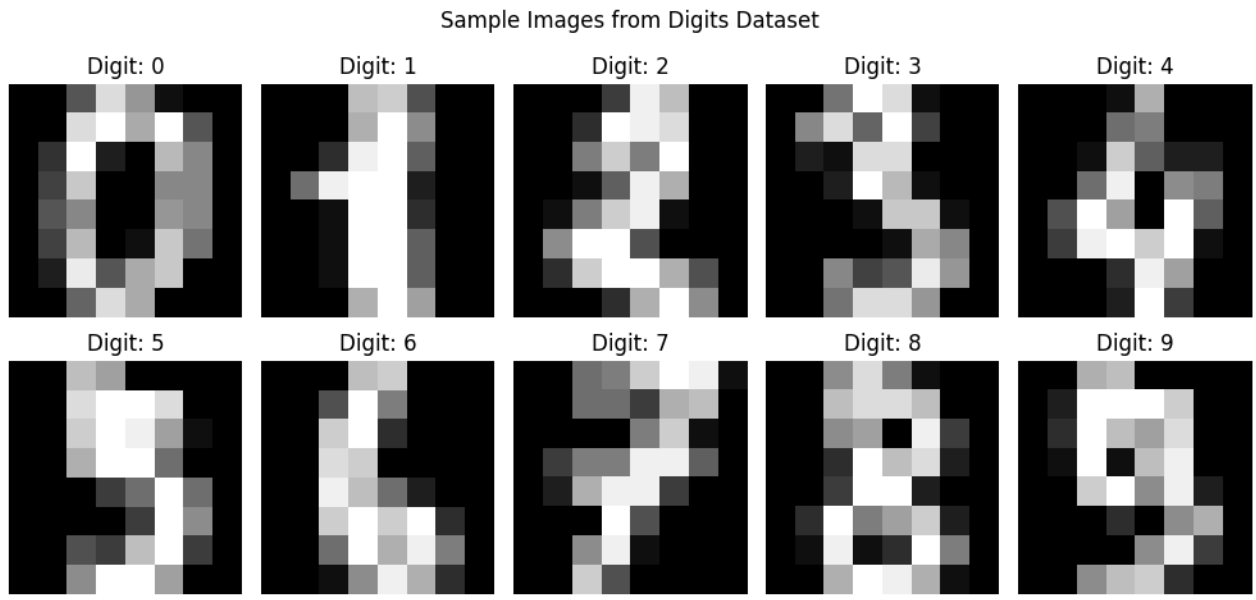
print("Dataset shape:", X.shape)
print("Number of images:", len(X))
print("Features per image:", X.shape[1])
```

```
Dataset shape: (1797, 64)
Number of images: 1797
Features per image: 64
```

```
fig, axes = plt.subplots(2, 5, figsize=(10, 5))

for i, ax in enumerate(axes.flat):
    ax.imshow(digits.images[i], cmap="gray")
    ax.set_title(f"Digit: {y[i]}")
    ax.axis("off")

plt.suptitle("Sample Images from Digits Dataset")
plt.tight_layout()
plt.show()
```




```
kmeans = KMeans(
    n_clusters=10,
    random_state=42,
    n_init=10
)

clusters = kmeans.fit_predict(X)

print("\nK-Means Clustering Results")
print("Number of clusters:", kmeans.n_clusters)
print("Inertia:", round(kmeans.inertia_, 2))
```

```
K-Means Clustering Results
Number of clusters: 10
Inertia: 1165256.3
```

```
silhouette = silhouette_score(X, clusters)
ari = adjusted_rand_score(y, clusters)
```

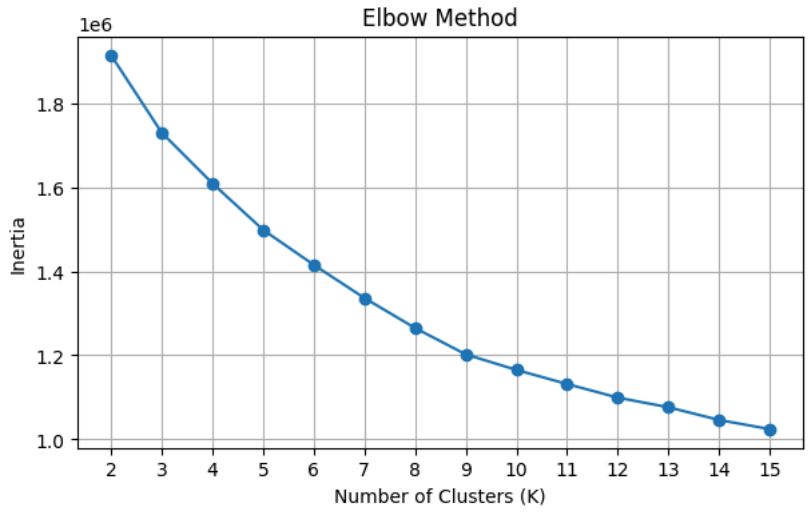
```
print("Silhouette Score:", round(silhouette, 4))
print("Adjusted Rand Index:", round(ari, 4))
```

```
Silhouette Score: 0.1824
Adjusted Rand Index: 0.6669
```

```
inertias = []

for k in range(2, 16):
    model = KMeans(
        n_clusters=k,
        random_state=42,
        n_init=10
    )
    model.fit(X)
    inertias.append(model.inertia_)
```

```
plt.figure(figsize=(7, 4))
plt.plot(range(2, 16), inertias, marker="o")
plt.title("Elbow Method")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Inertia")
plt.xticks(range(2, 16))
plt.grid(True)
plt.show()
```




```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)

centers_pca = pca.transform(kmeans.cluster_centers_)

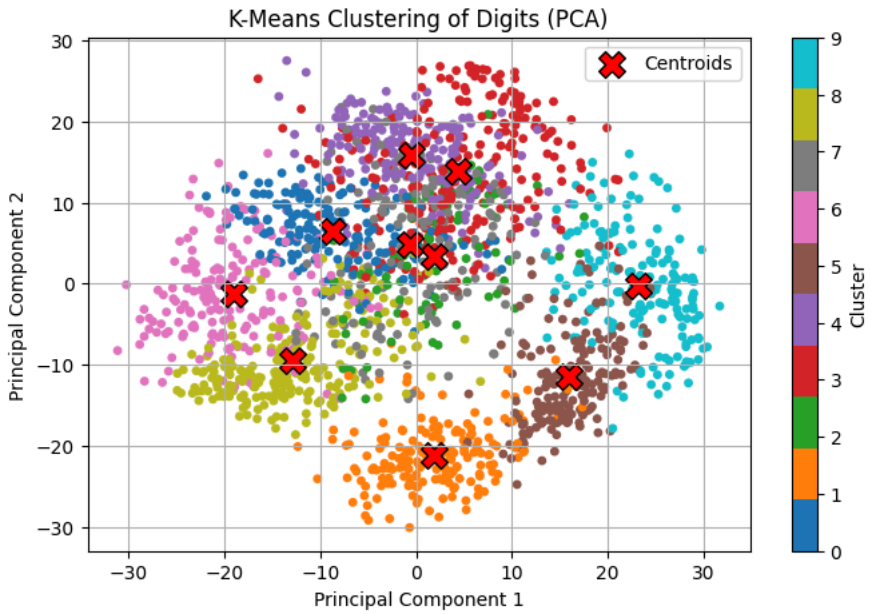
plt.figure(figsize=(8, 5))

scatter = plt.scatter(
    X_pca[:, 0],
    X_pca[:, 1],
    c=clusters,
    cmap="tab10",
    s=15
)

plt.scatter(
    centers_pca[:, 0],
    centers_pca[:, 1],
    marker="X",
    s=200,
    c="red",
    edgecolors="black",
    label="Centroids"
)

plt.title("K-Means Clustering of Digits (PCA)")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.legend()
plt.colorbar(scatter, label="Cluster")
plt.grid(True)
```

```
plt.show()
```




```
plt.figure(figsize=(8, 5))

scatter = plt.scatter(
    X_pca[:, 0],
    X_pca[:, 1],
    c=y,
    cmap="tab10",
    s=15
)

plt.title("Actual Digit Labels (PCA)")
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.colorbar(scatter, label="Actual Digit")
plt.grid(True)
plt.show()
```

